RetailPulse AI

End-to-End Retail Sales Forecasting & Business Intelligence Platform — Project Summary Report

1. Project Overview

RetailPulse AI is an end-to-end forecasting platform built on the Rossmann Store Sales dataset (1,017,209 daily records across 1,115 stores, January 2013 to July 2015). The project progressed through four phases: classical time-series forecasting, statistical ARIMA/SARIMA modeling, Prophet forecasting, and finally supervised machine learning enriched with store-level business features. The best-performing model, XGBoost Regression, was deployed as an interactive Streamlit application.

2. Data Understanding & EDA

- No missing values in the transaction-level train data; missing values in the store metadata (competition and Promo2 fields) reflect genuine absence of competitors or promotions, not data errors.
- Of 172,871 zero-sales records, 172,817 occurred while the store was closed — zero sales are overwhelmingly operational, not demand-driven.
- Strong weekly seasonality: Monday sales average 7,809 vs. a near-flat Sunday average of 204 (most stores closed).
- Active promotions lift average sales by ~81% (4,406 → 7,991); wider assortments and closer competitors also correlate with higher sales.
- Store Type B stores generate roughly double the average revenue of other store types.

3. Modeling Phases & Results

Phase 1 — Classical Forecasting

Aggregated daily sales were modeled with naive, moving-average, and exponential-smoothing methods. Holt-Winters Triple Exponential Smoothing (capturing level, trend, and weekly seasonality) was the best classical model.

Model	MAE	RMSE	MAPE
Holt-Winters (best)	1,250,170	1,789,503	88.84%
Seasonal Naive	2,911,575	4,072,885	119.21%
Naive Forecast	2,687,301	3,851,607	543.02%

Phase 2 — ARIMA & SARIMA

The series was confirmed stationary (ADF $p \approx 6.4\times10^{-5}$, $d = 0$). ACF/PACF analysis showed strong seasonal spikes at lags 7/14/21/28. SARIMA(1,0,1)(1,0,1,7), which explicitly models weekly seasonality, sharply outperformed non-seasonal ARIMA(1,0,1).

Model	MAE	RMSE	MAPE
ARIMA(1,0,1)	2,520,906	3,264,490	381.57%
SARIMA(1,0,1)(1,0,1,7) (best)	1,302,161	1,913,409	71.61%
SARIMA(5,0,0)(1,0,1,7)	1,507,593	2,128,380	86.62%

Phase 3 — Prophet Forecasting

Prophet automatically decomposed trend, weekly, and yearly seasonality without manual ACF/PACF tuning, but achieved MAE 1.51M / RMSE 2.10M / MAPE 107.32% — underperforming both Holt-Winters and SARIMA, confirming that explicit weekly-seasonality modeling was more effective for this dataset.

Phase 4 — Machine Learning (Store-Level Features)

Train and store metadata were merged (844,392 active, open-store records after filtering closures) and engineered into 24 features, including CompetitionAge and Promo2Age. XGBoost delivered the lowest error of any model tested.

Model	MAE	RMSE	MAPE
Mean Baseline	2,239.94	3,070.02	—
Linear Regression	1,991.76	2,761.12	42.31%
Random Forest	804.17	1,190.38	11.43%
XGBoost (deployed)	792.78	1,137.41	11.59%

Top predictive features (Random Forest importance): CompetitionDistance (20.1%), Store identity (16.6%), Promo (14.8%), competition-opening timing (~13%), DayOfWeek (6.5%), WeekOfYear (4.0%).

4. Deployment

The final XGBoost model and its LabelEncoders were serialized with Joblib and deployed inside a Streamlit application (RetailPulse AI) featuring an overview dashboard, phase-by-phase visual walkthroughs of every forecasting approach, and a live sales-prediction interface using the exact 21-feature training schema.

5. Business Insights & Recommendations

- Promotions are the single most controllable lever, lifting average sales by ~81% — promotional calendars should be optimized around this effect.
- Weekly demand cycles dominate behavior; staffing and inventory replenishment should be scheduled around the Monday peak and Sunday closures.
- Competitive proximity and store-specific characteristics matter more than calendar effects for absolute sales level, suggesting store-level (rather than purely global) forecasting refinements.
- Machine learning models that incorporate business context (competition, promotions, store attributes) substantially outperform pure time-series methods for day-level, store-level forecasting.

6. Conclusion

Across four modeling phases, incorporating richer information — first seasonality, then business context — consistently reduced forecasting error. XGBoost, leveraging store-level and promotional features, was selected as RetailPulse AI's production model, reducing MAE by over 99% relative to the mean baseline and by ~1.4% relative to Random Forest.